

Lezione 5 distanza + esercizi

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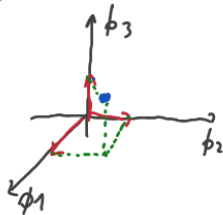
Vediamo ora il nesso tra funzioni e classi di funzioni ortonormali per trovare una base (facendo un riepilogo):

$$\{x_i(t)\}, \underbrace{\{\phi_k(t)\}}_{\substack{\text{classe di} \\ \text{funzioni} \\ \text{ortonormali}}}, (\phi_k, \phi_h) = \begin{cases} 0 & \text{se } h \neq k \\ 1 & \text{se } h = k \end{cases}$$

Se il sistema è completo si ha che:

$$x(t) = \sum_n a_n \phi_n(t), \text{ con } \underline{N \text{ finito}}$$

→ ϕ_n base



$$x(t) = \sum_n a_n \phi_n(t)$$

$$a_n = \int x(t) \phi_n^*(t) dt = (x, \phi_n)$$

Vediamo ora la distanza tra due segnali:

$$\int_0^T |x(t) - y(t)|^2 dt = \int_0^T [x(t) - y(t)] [x^*(t) - y^*(t)] dt$$

$$= \int_0^T \left[\sum_n a_n \phi_n(t) - \sum_h b_h \phi_h(t) \right] \left[\sum_i a_i^* \phi_i^*(t) - \sum_j b_j^* \phi_j^*(t) \right] dt$$

$$= \underbrace{\sum_n |a_n|^2}_{E_{xx}} + \underbrace{\sum_h |b_h|^2}_{E_{yy}} - \underbrace{\sum_n a_n^* b_n}_{\substack{\text{prodotto} \\ \text{scalare}}} =$$

$$E_{xx} + E_{yy} - 2 \operatorname{Re}(x, y)$$

La somma cambia il " - " in " + "

Facendo un discorso generale (usando la normalizzazione):

$$\{x_i\}, \{x_1(t), x_2(t), \dots, x_m(t)\}$$

$$\phi_1 = \frac{x_1(t)}{\sqrt{E_{x_1 x_1}}}$$

$$\phi_2 = \frac{x_2(t) - (x_2, \phi_1) \phi_1(t)}{\sqrt{E_{x_2 x_2}}}$$

$$= (x_2, \phi_1) \phi_1(t) = 0$$

$$= \frac{x_2(t) - (x_2, \phi_1) \phi_1(t)}{\sqrt{E_{x_2 x_2}}}$$

$$\phi_3 = x_3(t) - (x_3, \phi_1) \phi_1(t) - (x_3, \phi_2) \phi_2(t)$$

$$= [x_3(t) - (x_3, \phi_1) \phi_1(t) - (x_3, \phi_2) \phi_2(t)]$$

$$= \frac{x_3(t) - (x_3, \phi_1) \phi_1(t)}{\sqrt{E_{x_3 x_3}}}$$

1 + 2 + ...

Vediamo ora un po' di esercizi:

1) $(x, y)(\tau) = \int x(t-\tau) y^*(t) dt$, $y(t) = e^{-t}$ $t > 0$ *am*
 $x(t) = e^{-2t}$ $0 < t < 2$ *am*
 $0 < t+\tau < 2 \Rightarrow t > -\tau$
 $t < 2-\tau$

2) $(x, y)(\tau) = \int x(t+\tau) y^*(t) dt$, $x(t) = e^{-t}$ $t > 0$ *am*
 $y(t) = e^{-2t}$ $0 < t < 2$ *am*
 $t+\tau > 0$
 $t > -\tau$

$\Rightarrow -\tau < 0 \Rightarrow \tau > 0 \Rightarrow \int_0^2 e^{-(t+\tau)} e^{-2t} dt$ *1° caso*

$0 < \tau < 2 \Rightarrow \begin{cases} -\tau > 0 \Rightarrow \tau < 0 \\ -\tau < 2 \Rightarrow \tau > -2 \end{cases} \Rightarrow \int_{-\tau}^2 \dots$

$\Rightarrow -\tau > 2 \Rightarrow \tau < -2 \Rightarrow \int_{-\tau}^2 \dots$ *3° caso (integrale nullo)*

Facciamo ora l'integrale:

$\int e^{-(t+\tau)} e^{-2t} dt = e^{-\tau} \int e^{-3t} dt = e^{-\tau} \left[\frac{e^{-3t}}{-3} \right]_b^a$

3) $\mathcal{F}\{(x, y)(\tau) = \int x(t) y^*(t) dt\}$, $x(t) = e^{-t} u(t) \Rightarrow \frac{1}{1+j2\pi f}$

$y(t) = e^{-2t} u(t)$

$\int_0^\infty e^{-t} e^{j2\pi f t} dt = \left[\frac{e^{-(1-j2\pi f)t}}{-(1-j2\pi f)} \right]_0^\infty$

$\Rightarrow \frac{1}{1-j2\pi f}$

$\Rightarrow \frac{1}{(1+j2\pi f)(1-j2\pi f)} = \text{Frazioni parziali: } \frac{A}{1+j2\pi f} + \frac{B}{1-j2\pi f}$

$\Rightarrow \frac{A}{1+j2\pi f} + \frac{B}{1-j2\pi f} [1 - 2^{-4} e^{-j4\pi f}] = e^{-t} u(t) - 2^{-t} u(t) e^{-4}$

$\Rightarrow e^{-t} u(t) e^{-4} = e^{-4} u(t-2) [A] \text{ e } 2^{-t} u(t) e^{-4} = e^{-4} u(t-2) [B]$

4) $\mathcal{F}\left[\frac{1}{t^2-1}\right] = \frac{1}{t^2-1} = \frac{1}{(t-1)(t+1)} = \frac{A}{t-1} + \frac{B}{t+1}$

$A(t+1) + B(t-1) = 1 \Rightarrow \begin{cases} A+B=1 \\ A-B=1 \end{cases} \Rightarrow \begin{cases} A=1 \\ B=0 \end{cases}$

[B]

$\frac{1}{t^2-1} \Rightarrow e^{-2\pi f t} \Leftrightarrow \frac{2\omega}{d^2 + \omega^2} \Leftrightarrow \frac{2\omega}{d^2 + \omega^2}$

$\Rightarrow \beta e^{-\alpha|f|} \Leftrightarrow \frac{2\omega\beta}{\omega^2 + \alpha^2} \text{ ma } \frac{1}{2\pi} = 1$
 $\left(\frac{1}{2\pi}\right)^2 \omega^2 \Rightarrow \boxed{\alpha = 2\pi}$

$B = \frac{1}{2\pi}$

$\Rightarrow B \pi e^{-2\pi|f|} = \frac{B}{1+t^2}$

[A] $\frac{A}{t^2-1} = \frac{A_1}{t+1} + \frac{A_2}{t-1} = A_1 t - A_1 + A_2 t + A_2 = A \Rightarrow \begin{cases} A_1 = \frac{A}{2} \\ A_2 = \frac{A}{2} \end{cases}$

$\mathcal{F}\{\sin(\pi t)\} = \frac{1}{j\pi f} \Leftrightarrow \sin(\pi f) = \frac{1}{j\pi f}$
 $\Rightarrow \sin(\pi f) = \frac{1}{j\pi f}$

$(-j\pi) \sin(\pi f) \Leftrightarrow \frac{1}{f}$

$\Rightarrow (-j\pi) \sin(\pi f) \left[\frac{e^{-j2\pi f} + e^{j2\pi f}}{2j} \right] = \pi \sin(\pi f) \sin(2\pi f)$

$$5) y''(t) = y(t) + f(t) \Rightarrow (j\omega)^2 Y(f) = Y(f) + 1$$

$$= -\omega^2 Y(f) - Y(f) = -1$$

$$= Y(f) = \frac{-1}{-\omega^2 - 1}$$

homogene

$$y''(t) = y(t) \Rightarrow e^{\lambda t} = e^{\lambda t} \Rightarrow \lambda^2 = 1 \Rightarrow \lambda = \pm 1$$

$$\Delta e^t + \beta e^{-t} \Rightarrow e^t u(t) + e^{-t} u(t) \Rightarrow y(t) = \frac{e^{-|t|}}{2}$$

$$y'(t) = -\frac{e^{-|t|}}{2} \text{ sign}(t)$$

$$y''(t) = \frac{e^{-|t|}}{2} \text{ sign}(t) \text{ sign}(t) = \frac{e^{-|t|}}{2} \checkmark f(t)$$

$$\Rightarrow \frac{e^{-|t|}}{2} + e^{|t|} f(t) = \frac{e^{-|t|}}{2} f(t)$$

$$6) x(t) = \text{erfc}(t) \text{ limits to } -\frac{\pi}{2} \text{ to } \frac{\pi}{2}$$

$$\frac{dx(t)}{dt} = \frac{1}{t^2 + 1} \Rightarrow \pi e^{-\pi f} \left[\text{erfc}(t) = \frac{\pi}{2} \text{ sign}(t) \right]$$

$$j2\pi f X(f) = \frac{1}{e^{-\pi f^2}} \Rightarrow X(f) = \frac{e^{-\pi f^2}}{j2\pi f}$$

$$7) x(t) = \left[\frac{1}{t^2 + 1} \right] \Rightarrow \int_{-\infty}^{\infty} \frac{1}{(t^2 + 1)^2} dt = \frac{1}{2} \left[\frac{t}{t^2 + 1} + \text{erfc}(t) \right] \Big|_{-\infty}^{\infty} \text{ difficile}$$

$$\Rightarrow \frac{1}{\pi} \int_{-\infty}^{\infty} e^{-\pi f^2} |x(f)|^2 df$$

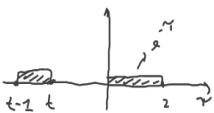
$$= \frac{2}{\pi^2} \int_{-\infty}^{\infty} e^{-\pi f^2} df$$

$$9) \text{ LTI } x(t) = t, 0 < t < 1$$

$$h(t) = e^{-t}, 0 < t < 2$$

$$x(t) \cdot h(t) + y = \int (t-\tau) e^{-\tau} d\tau$$

$$0 < t-\tau < 1 \Rightarrow \begin{cases} \tau < t \\ \tau > t-1 \end{cases}$$



1) $t < 0$ solution

$$2) \begin{cases} t > 0 \\ t-1 < 0 \end{cases} \Rightarrow 0 < t < 1 \Rightarrow \int_0^t x(\tau) h(t-\tau) d\tau$$

$$3) \begin{cases} t < 2 \\ t-1 > 0 \end{cases} \Rightarrow 1 < t < 2 \Rightarrow \int_{t-1}^t x(\tau) h(t-\tau) d\tau$$

$$4) \begin{cases} t > 2 \\ t-1 > 2 \end{cases} \Rightarrow 2 < t < 3 \Rightarrow \int_{t-2}^t x(\tau) h(t-\tau) d\tau$$

$$5) t > 3 \Rightarrow 0$$

$$10) x(t) = t \Rightarrow \mathcal{F}[x(t)] = \int_{-\infty}^{\infty} t e^{-j\omega t} dt = \mathcal{F}[t] = \frac{j}{\omega^2}$$

$$\begin{aligned} \mathcal{F}[t] &\leftrightarrow \frac{j}{\omega^2} \\ \mathcal{F}[t^2] &\leftrightarrow \frac{j^2}{\omega^3} \\ \mathcal{F}[t^3] &\leftrightarrow \frac{j^3}{\omega^4} \end{aligned}$$

$$\Rightarrow \frac{1}{(-j\omega)^2} = \frac{1}{-\omega^2} = -\frac{1}{\omega^2}$$

$$\Rightarrow \mathcal{F}[x(t)] = \mathcal{F}[t] = \frac{j}{\omega^2}$$

$$\Rightarrow \mathcal{F}[x(t)] = \frac{j}{\omega^2}$$

$$11) x(t) = t, t > 0 \Rightarrow t u(t) \Rightarrow \mathcal{F}[t u(t)] = \frac{1}{j\omega^2} + \frac{1}{j\omega}$$

$$= \frac{1}{j\omega^2} \left[-\frac{1}{j\omega^2} + \frac{f'(t)}{2} \right]$$

oppre:

$$x(t) + t u(t) \Rightarrow \frac{d x(t)}{dt} = u(t) + t \delta(t)$$

$$\frac{d x(t)}{dt} = u(t) \Rightarrow \int_{-\infty}^{\infty} \frac{d x(t)}{dt} = \int_{-\infty}^{\infty} u(t) = \frac{1}{j2\pi f} + \frac{\delta(f)}{2}$$

$$\Rightarrow x(f) = \frac{1}{(j2\pi f)} \left[\frac{1}{j2\pi f} + \frac{\delta(f)}{2} \right]$$

$$11) x(t) = |t| \quad \forall t \Rightarrow t \text{ sign}(t) = \frac{1}{-2j\pi} \frac{d}{df} \left(\frac{1}{j2\pi f} \right) = - \frac{1}{(j2\pi)(j2\pi f^2)}$$

$$= \frac{-1}{2\pi f^2}$$

oppre:

$$\frac{d x(t)}{dt} = \text{sign}(t) + t \delta(t) = (j2\pi f) x(f) + \frac{1}{j2\pi f}$$

$$13) x(t) = \frac{t}{t^2 + 1} \Rightarrow \frac{d}{df} = \frac{1}{2\pi} \quad ; \quad \frac{2\omega}{\omega^2} = \frac{2 \cdot 2\pi B}{\omega^2} = 1$$

$$\Rightarrow \pi e^{-2\pi|f|}$$

$$\Rightarrow \frac{1}{(-j2\pi)} \frac{d}{df} \left[\pi e^{-2\pi|f|} \right] = \frac{(-2\pi) \pi e^{-2\pi|f|}}{(-j2\pi)} \text{sign}(f)$$